

Pearson Edexcel International GCSE ICT (4IT1)

Paper 1: Impact of the Internet Mock Examination

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Total Marks: 150 Marks

Question 1 (Total: 30 marks)

The expansion of global telecommunications infrastructure has fundamentally altered how digital information and online services are managed.

(a) State the technical definition of the **internet**. (1)

(b) Differentiate between the internet and the World Wide Web by outlining the key distinction between devices and content. (2)

(c) Networked communications rely on standard rule sets known as protocols to translate data paths.

Identify the correct **generic application focus** or target service from the textbook for the following communication protocols:

(6)

• IMAP:

• FTP:

• SSH:

• VoIP:

• XMPP:

• HTTP:

(d) Interactivity on the web has transitioned through distinct evolutionary phases.

(i) Define the term **Web 2.0**. (2)

(ii) State what is meant by an **internet meme**. (1)

(e) Identify one activity from the list below that is uniquely possible only when using an active internet connection. Place a cross (x) in the correct box. (1)

☐

A. Viewing analogue television broadcasts

☐

B. Listening to a local FM radio receiver

☐

C. Media streaming of online video content

☐

D. Placing offline mobile telephone voice calls

(f) The internet provides individuals with rapid entry channels to public information networks.

State three examples of services or systems that users can access or utilize via internet connectivity. (3)

1.

2.

3.

(g) The introduction of internet-centric workflows alters corporate expectations regarding staff competencies.

(i) Explain why the internet introduces new skill requirements for existing workforces. (2)

(ii) Explain how a failure to regularly update these technical skills can lead to structural job losses. (2)

(iii) List two new job opportunities created by the progression of internet systems and digital architectures. (2)

1.

2.

(h) Describe how the internet accelerates the spread of public data (such as news, sports, or weather alerts) across the globe. (2)

(i) State three distinct areas where skilled technology workers are needed to advise, build, or deploy digital architectures for corporate operations. (3)

1.

2.

3.

Question 2 (Total: 30 marks)

Modern organizations deploy diverse working models, including collaborative setups, flexible pathways, and remote desktop environments.

(a) State the definition of **collaborative working**. (1)

(b) When organizing a collaborative project framework, management can choose between splitting workflows into narrow individual components or assigning multiple operators to identical tracks.

Complete the table below by outlining the exact textbook benefits and drawbacks of each strategy. (8)

Collaboration Strategy	Sourced Operational Benefit	Sourced Operational Drawback
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Splitting into narrow individual tasks	(i)	(ii)
Assigning more than one person to a task	(iii)	(iv)

(c) Differentiate between flexible working and mobile working configurations in terms of employee location and scheduling properties. (4)

Flexible Working:

Mobile Working:

(d) Review the structural parameters of non-traditional employment routines.

(i) State two benefits to an individual employee of adopting a flexible working plan. (2)

1.

2.

(ii) State two drawbacks to an individual employee that can result from poorly managed mobile working routines. (2)

3.

4.

(e) Many corporate agencies transition their staff towards permanent online work-from-home models.

(i) Explain two distinct benefits experienced by individuals who switch to working from home online. (4)

Benefit 1:

Benefit 2:

(ii) Explain two distinct benefits gained by organizations that dismantle physical offices in favor of remote operations. (4)

Benefit 1:

Benefit 2:

(iii) State three drawbacks that remote working models impose on an organization's management or administrative overheads. (3)

1.

2.

3.

(f) Identify the core psychological drawback an individual face-to-face employee might experience if isolated from colleagues over long remote production periods. (2)

Question 3 (Total: 35 marks)

The presence of global internet portals reshapes data safety, tracking metrics, and defense requirements.

(a) Define the term **digital footprint**. (1)

(b) Explain why user-generated content or social media updates can pose a permanent privacy risk to individuals over extended periods. (2)

(c) Educational safety initiatives use specific mnemonic frameworks like "Zip It, Block It, Flag It" to train young users. Describe the precise action routine mandated by each directive below: (6)

Zip It:

Block It:

Flag It:

(d) State four specific types of private personal information that an individual should never share with strangers online. (4)

1.

2.

3.

4.

(e) Organizations utilize the internet to achieve optimization milestones and uncover behavioral insights.

(i) Describe how the internet improves communication channels between corporate organizations, their employees, and customer networks. (2)

(ii) Explain how internet portals grant organizations frictionless access to global markets without physical logistical charges. (2)

(iii) State two advantages an organization gains by hiring from a global workforce distributed across multiple countries. (2)

1.

2.

(e)[ii] Identify the technical term used to describe the collection and analysis of massive quantities of data harvested from diverse sources. (1)

(ii)[alt] State the primary objective of tracking and processing these large data blocks. (1)

(f) Expanding remote connections can compromise corporate network boundaries.
Explain three specific methods by which unauthorized users or hackers target and exploit remote organization structures to steal core data records. (6)

Method 1:

Method 2:

Method 3:

(g) The emergence of digital market alternatives creates substantial competition that can threaten entire legacy sectors. Identify four distinct legacy industries or business models that have been disrupted or put at financial risk by internet-based alternatives. (4)

1.	2.
3.	4.

(h) Explain why cyberbullying can cause greater distress to an individual than traditional in-person bullying, highlighting properties of speed and identity concealment. (4)

Question 4: Technical Comparison Matrix (Total: 30 marks)

An IT sociologist is preparing an infrastructure evaluation checking the digital divide across global populations and evaluating changes in public network usage.
Complete the missing properties within the classification tables below by directly sourcing technical criteria and definitions from the textbook guidelines. (30 marks, 5 marks per matrix cell)

Part A: Evaluating the Causes of Unequal Access to ICT

Divide Core Dimension	Sourced Technical Cause / Performance Criteria	Sourced Real-World Example
Economic & Infrastructure Blocks	Developing nations prioritize vital life support needs over technological installations.	Prioritizing food networks and healthcare arrays.
Physical Location Obstacles	(a)	Remote, rural, heavily forested, or mountainous areas.

Political Censorship Constraints	(b)	Governments deliberately blocking or disrupting internet access to control citizens.
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Part B: Defining Information Volumetrics and Structural Staging

Technical Category	Core Operational Feature / Impact Criteria	Sourced Social Status Profile
Information Rich	(c)	Enhances levels of education and improves decision-making paths.
Information Poor	(d)	Decreases access to education and reduces decision-making capabilities.
User-Generated Content	Reference portals challenge traditional concepts of verification because entries are not audited by a centralized panel.	(e)
Digital Divide	(f)	The gap separating people with device access from those without it.

Question 5: Extended Response Essay (Total: 25 marks)

A national education authority is planning to migrate its entire regional school administration and curriculum delivery to an internet-integrated cloud platform. The platform features Virtual Learning Environments (VLEs), accommodates asynchronous teacher training, allows real-time collaborative grading between departments, and uses big data processing to track student performance trends.

The technology board is debating between two structural implementation strategies:

- **Option A:** Deploying a completely unrestricted, interactive web portal that maximizes open communication channels, allows students to contribute user-generated reference entries, permits mobile working for instructors from any cellular node, and leverages public social media APIs to share school news.
- **Option B:** Enforcing a highly restricted private network model that limits access via strict firewalls, blocks inbound and outbound personal communication streams, permits only fixed-location access from inside physical campus facilities, and relies entirely on centralized experts to verify school reference databases.

Evaluate both options. You should discuss the impacts of each option on data confidentiality (including vulnerability to external hacking), educational collaboration efficiency, student safety online (preventing exposure to cyberbullying or malicious links), and the potential to worsen or bridge the digital divide for students from lower-income backgrounds. (25)

Plan your answer carefully. Marks will be awarded for structured layout, correct usage of ICT terms from the text, balanced evaluation arguments, and an effective final conclusion.

